

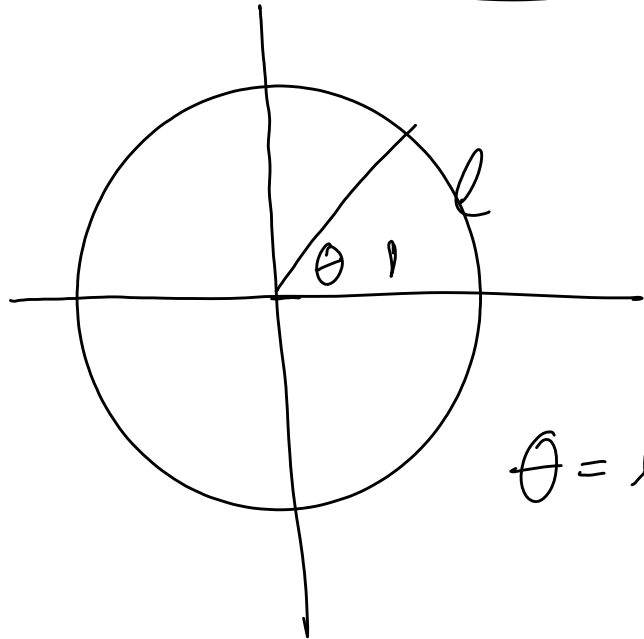
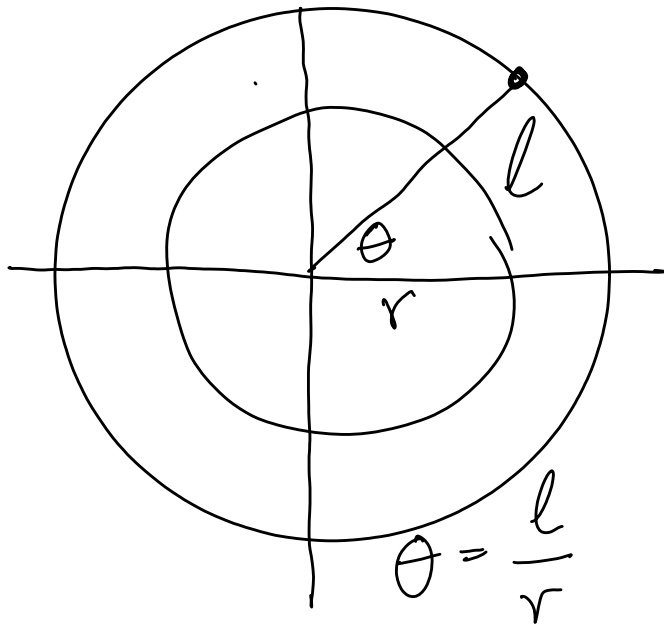
Unit 1.4 Trigonometry

A24 - A33

Note Title

2015/02/01

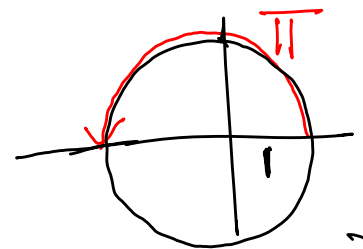
Radian measure



$$\theta = l.$$

Conversion:

$180^\circ = \pi$



May I have a large container of coffee?
3, 1 4 1 5 9 2 6

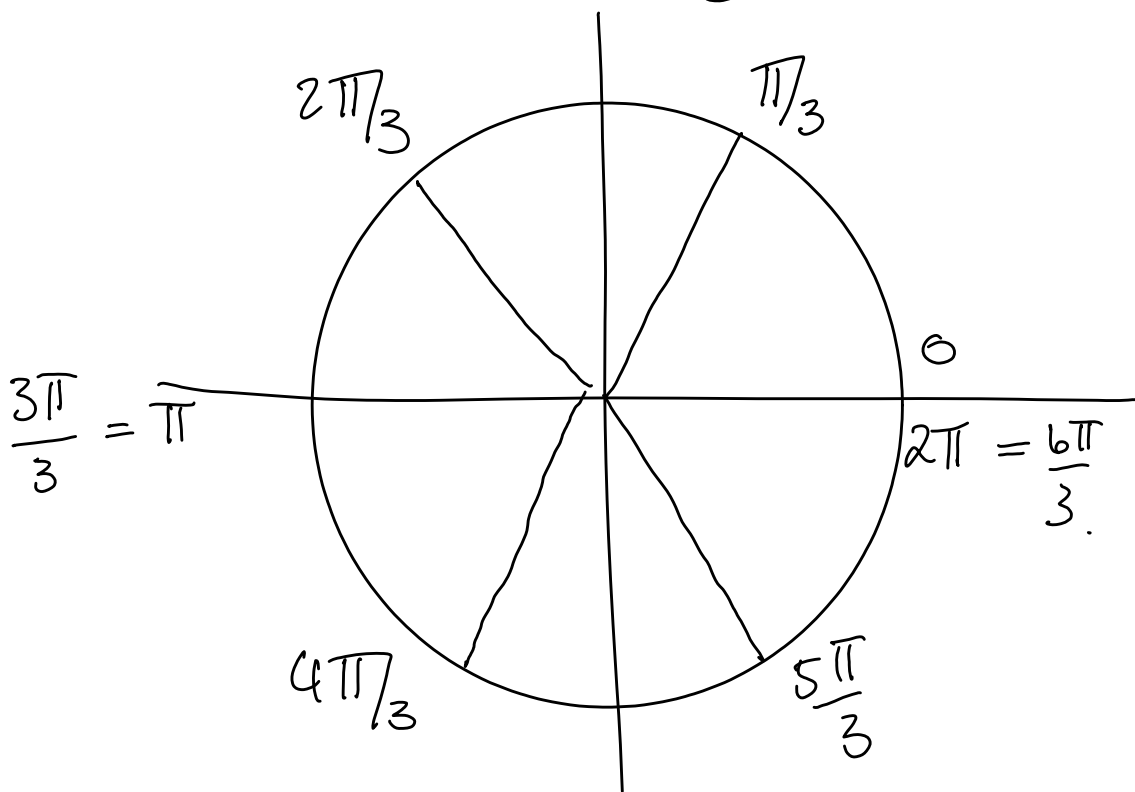
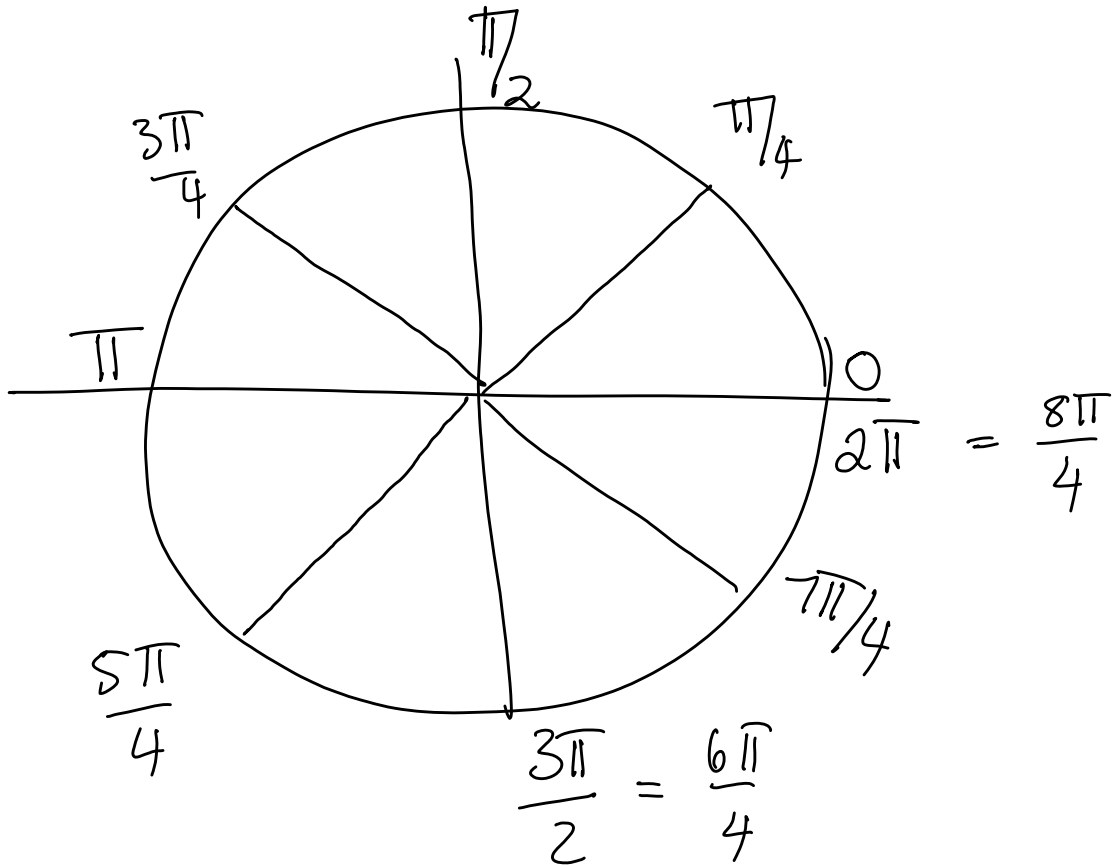
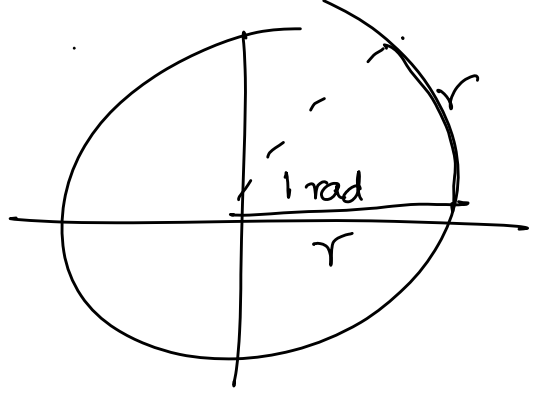
So:

360°	$= 2\pi$
180°	$= \pi$
90°	$= \frac{\pi}{2}$
60°	$= \frac{\pi}{3}$
45°	$= \frac{\pi}{4}$
30°	$= \frac{\pi}{6}$

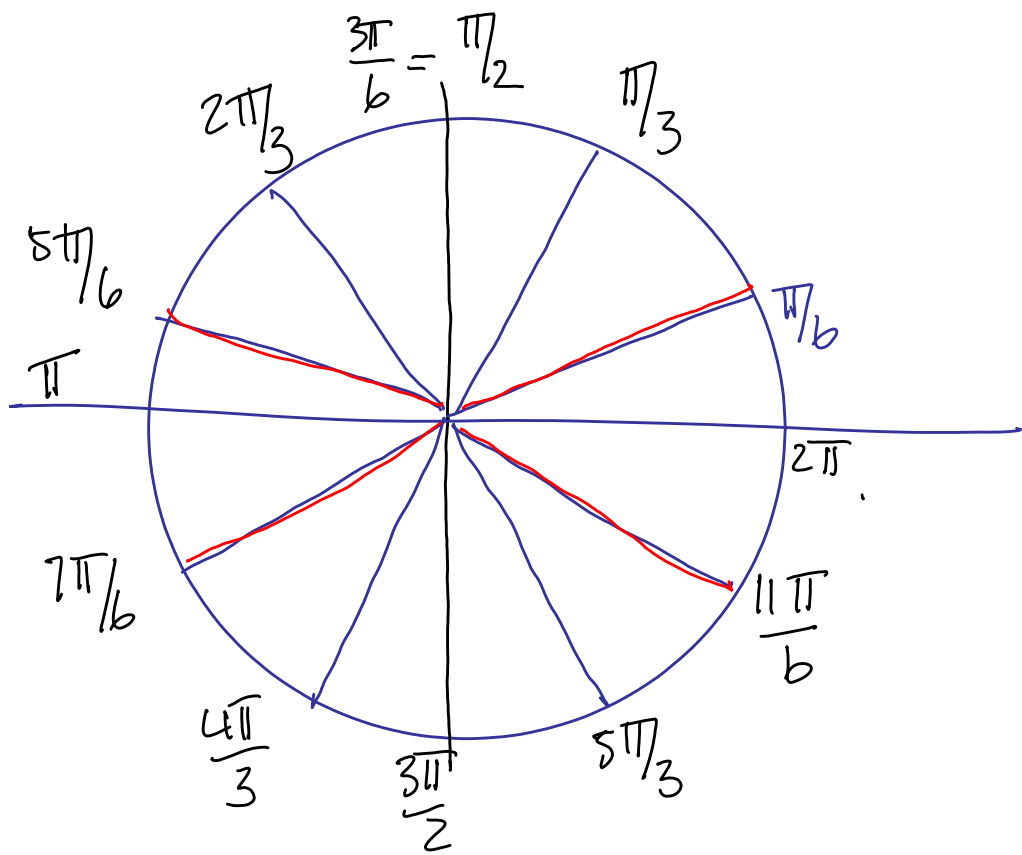
How big is 1 radian?

$$180^\circ = \pi \text{ rad.}$$

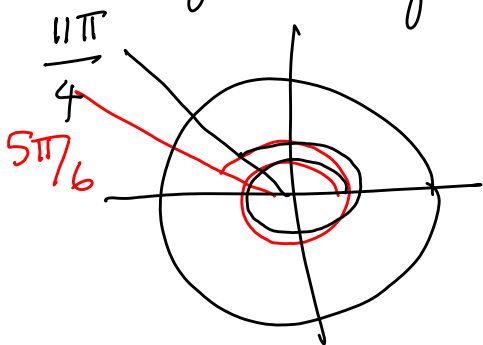
$$\frac{180^\circ}{\pi} = 1 \text{ rad} \\ \approx 57^\circ$$



$$\frac{6\pi}{b} =$$

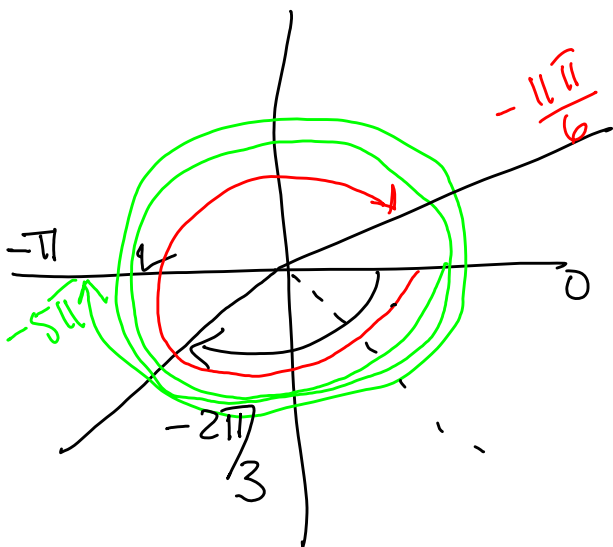


Angles greater than 2π or negative.



$$\frac{11\pi}{4} = \frac{8\pi}{4} + \frac{3\pi}{4}$$

$$\frac{17\pi}{6} = \frac{12\pi}{6} + \frac{5\pi}{6}$$

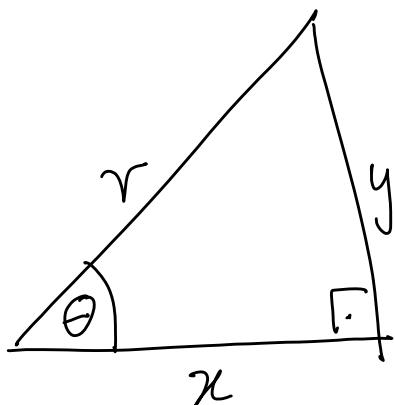


$$-\frac{2\pi}{3}$$

$$-\frac{11\pi}{6}$$

$$-5\pi$$

∴ The six trigonometric functions



$$\sin \theta = \frac{y}{r}$$

$$\cos \theta = \frac{x}{r}$$

$$\tan \theta = \frac{y}{x}$$

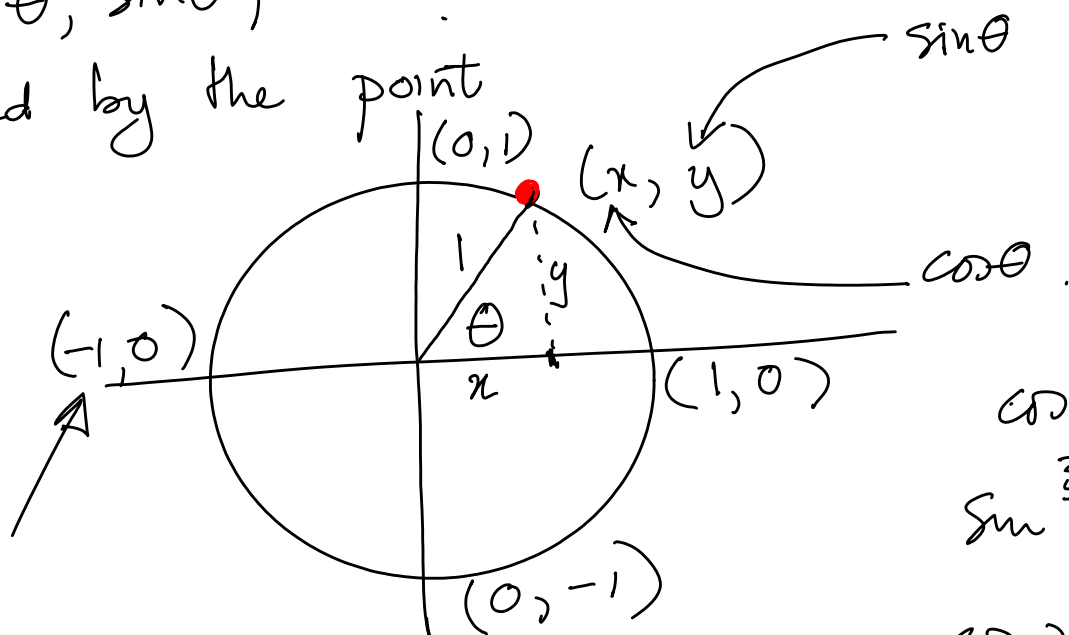
$$\operatorname{cosec} \theta = \frac{1}{\sin \theta} = \frac{r}{y}$$

$$\sec \theta = \frac{1}{\cos \theta} = \frac{r}{x}$$

$$\cot \theta = \frac{1}{\tan \theta} = \frac{x}{y}$$

In a unit circle:

A point (x, y) on the circle is $(\cos \theta, \sin \theta)$ where θ is the angle defined by the point



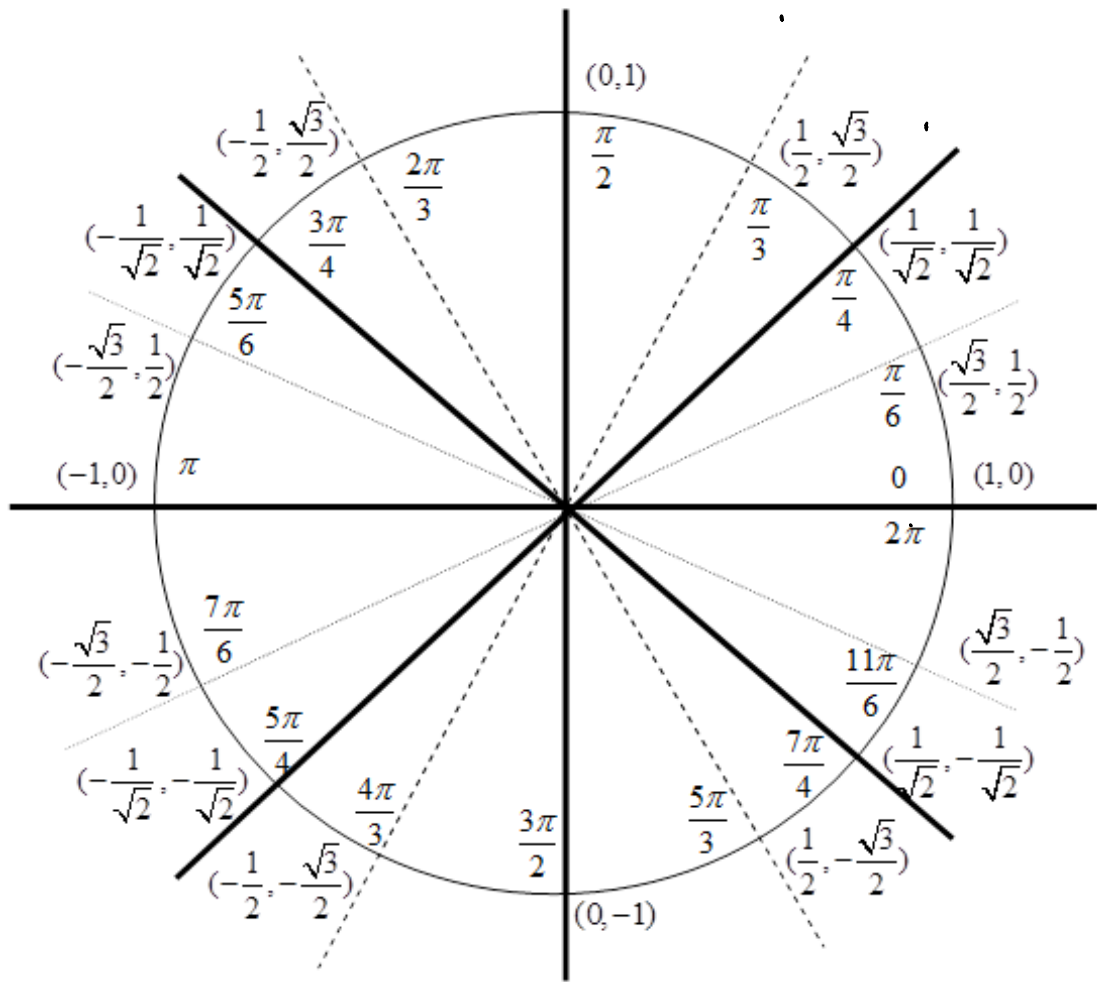
$$\cos \pi = -1$$

$$\sin \frac{3\pi}{2} = -1$$

$$\cos 2\pi = 1$$

$$\sin \frac{\pi}{2} = 1$$

The circle with sin/cosine values:



± 1
 $\pm \frac{1}{2}$
 $\pm \frac{1}{\sqrt{2}}$
 $\pm \frac{\sqrt{3}}{2}$

Find:

$$\sin \frac{\pi}{3} = \frac{\sqrt{3}}{2}$$

$$\cos \frac{2\pi}{3} = -\frac{1}{2}$$

$$\tan \frac{3\pi}{4} = -1$$

$$\sin \frac{7\pi}{6} = -\frac{1}{2}$$

$$\cos \frac{11\pi}{6} = \frac{\sqrt{3}}{2}$$

$$\sin \frac{13\pi}{6} = \frac{1}{2}$$

Solve for $\theta \in [0, 2\pi)$:

$$\sin \theta = -\frac{1}{2} \Rightarrow \theta = \frac{7\pi}{6} \text{ or } \theta = \frac{11\pi}{6}$$

$$\cos \theta = 1 \Rightarrow \theta = 0$$

$$\cos \theta = -\frac{1}{\sqrt{2}} \Rightarrow \theta = \frac{3\pi}{4} \text{ or } \theta = \frac{5\pi}{4}$$

$$\begin{aligned} \operatorname{cosec} x &= -\sqrt{2} \\ \Rightarrow x &= 5\pi/4 \text{ or } x = 7\pi/4 \quad \left| \sin x = -1/\sqrt{2} \right. \\ \cot x &= -1 \\ \Rightarrow x &= 3\pi/4 \text{ or } x = \pi/4 \\ \sec x &= 1 \\ \Rightarrow x &= 0. \end{aligned}$$

Solving trig equations:

For what value of x is

$$1. \cos 2x = \frac{1}{2}$$

$$\Rightarrow 2x = \pi/3 + 2k\pi \quad \text{or} \quad 2x = 5\pi/3 + 2k\pi,$$

$$\Rightarrow x = \pi/6 + k\pi \text{ or } x = 5\pi/6 + k\pi \quad \left| k \in \mathbb{Z} \right.$$

$$2. \cos^2 x - 2 \sin x - 1 = 0$$

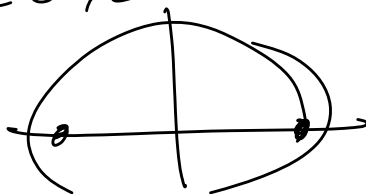
$$\Rightarrow 1 - \sin^2 x - 2 \sin x - 1 = 0.$$

$$\Rightarrow \sin^2 x + 2 \sin x = 0.$$

$$\Rightarrow \sin x (\sin x + 2) = 0$$

$$\Rightarrow \sin x = 0 \text{ or } \sin x = -2$$

$$\Rightarrow \underline{x = k\pi}, \quad k \in \mathbb{Z} \quad \text{N/A}$$



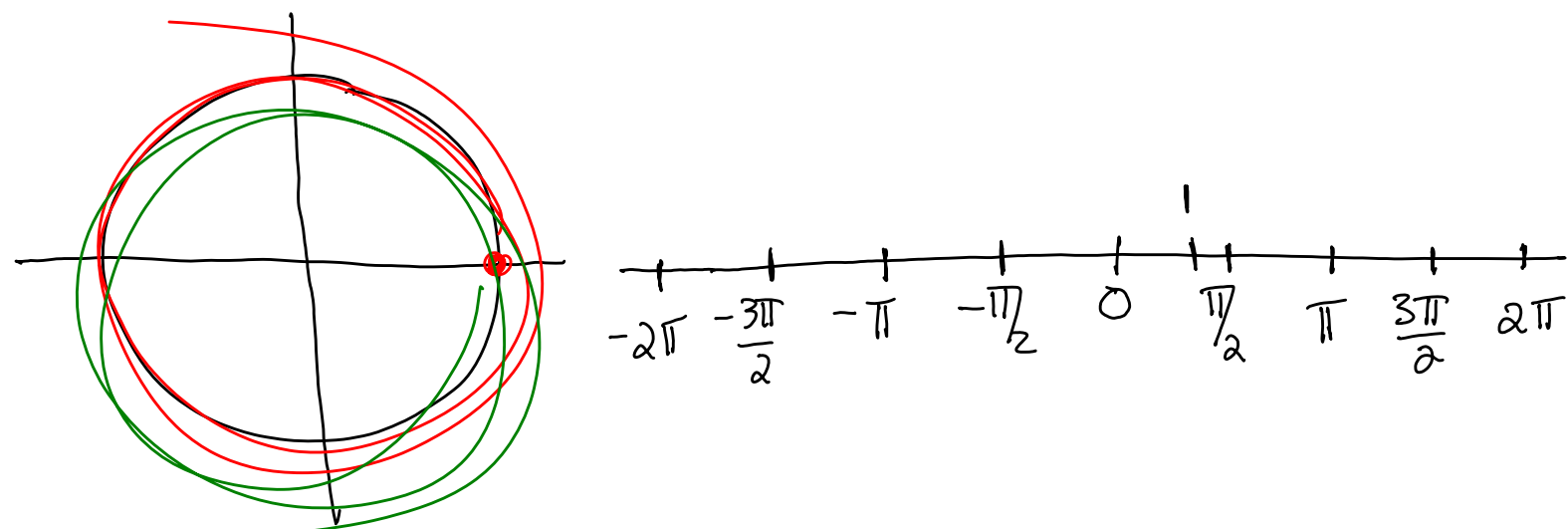
Identities

$$\sin^2 x + \cos^2 x = 1$$

$$\sec^2 x = 1 + \tan^2 x$$

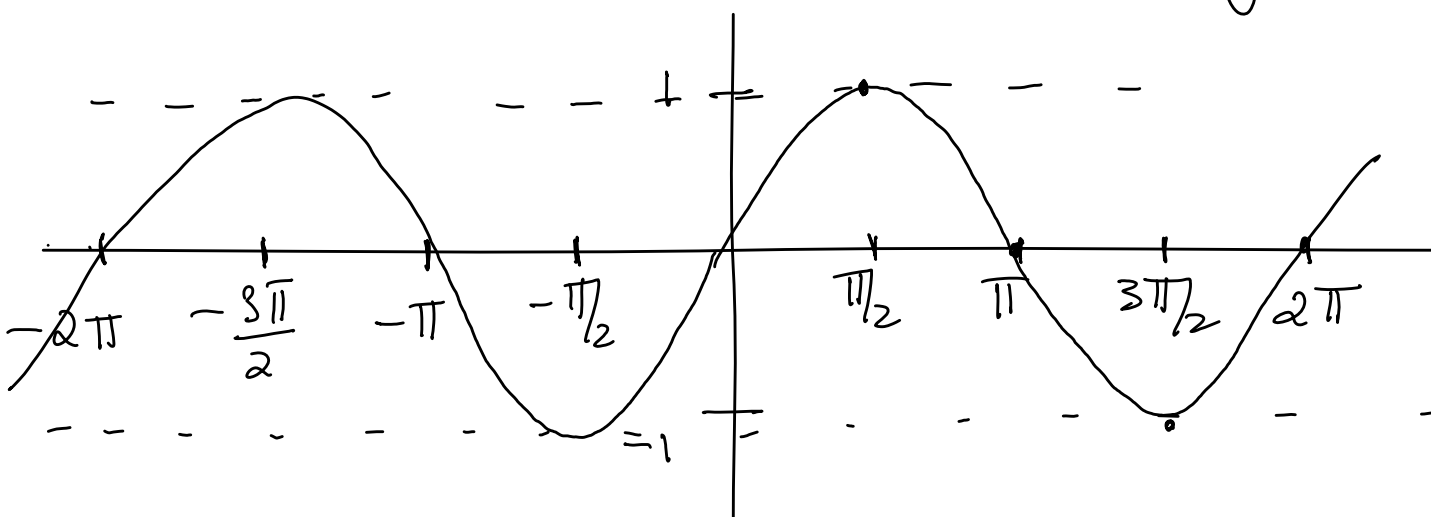
$$\operatorname{cosec}^2 x = 1 + \cot^2 x$$

From the circle to the real line

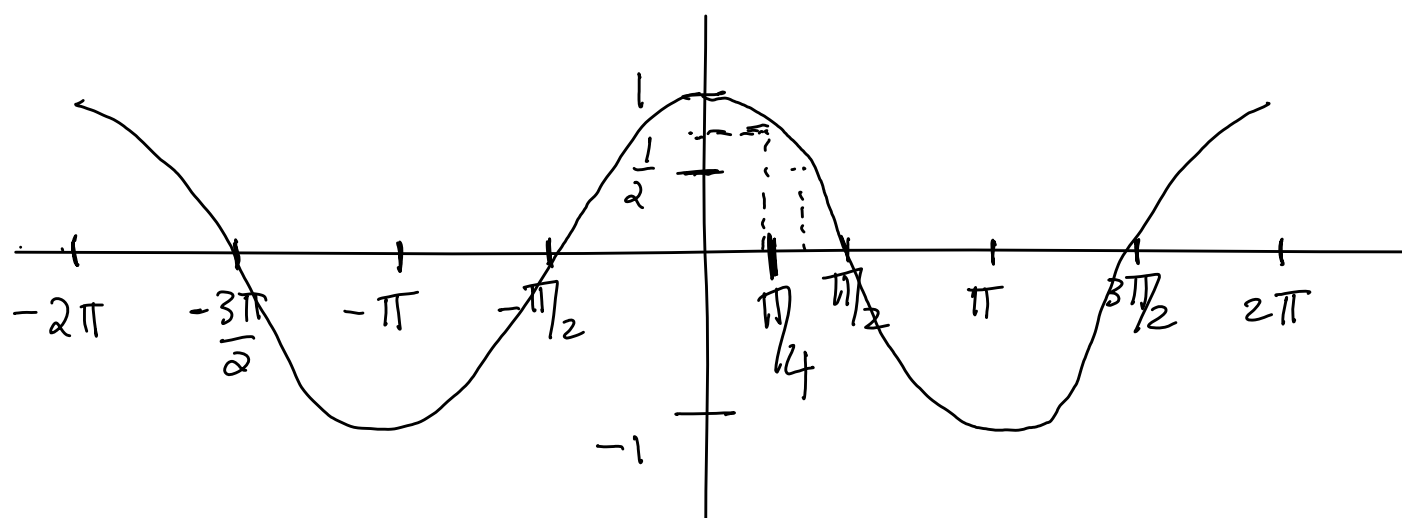


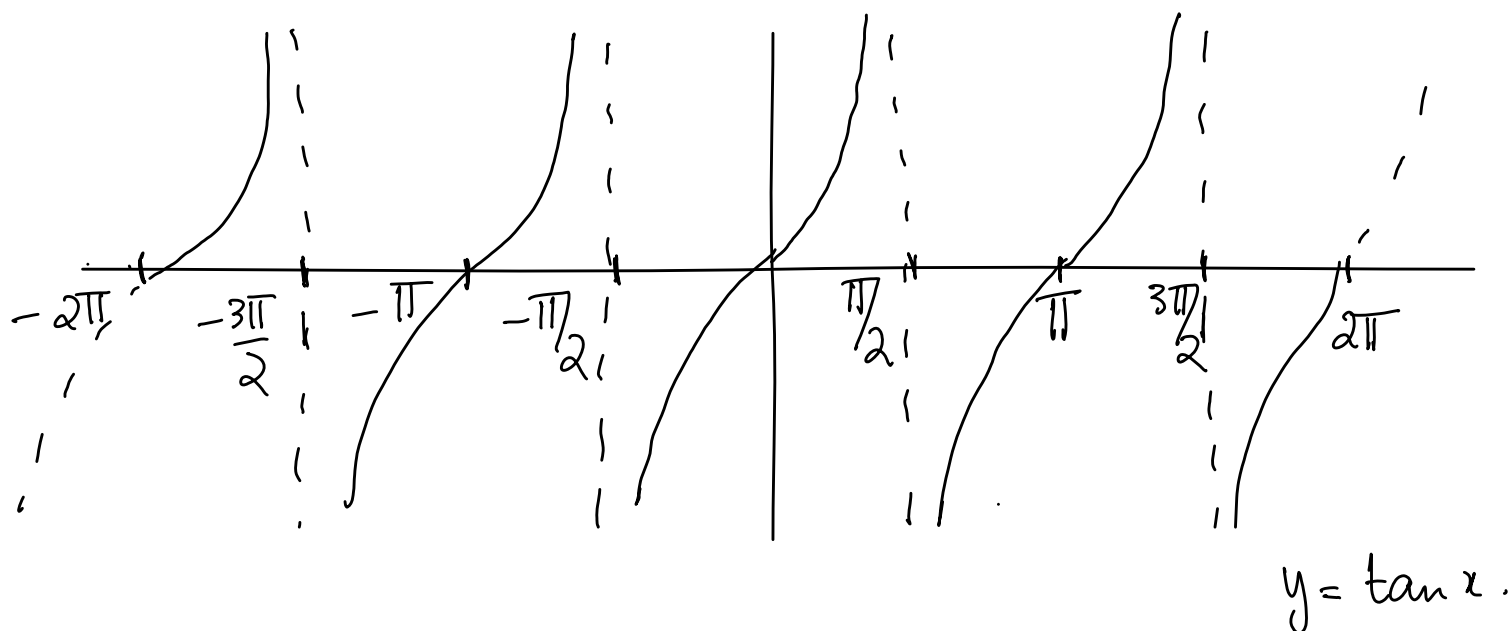
Graphs of the trig functions

$$y = \sin x$$



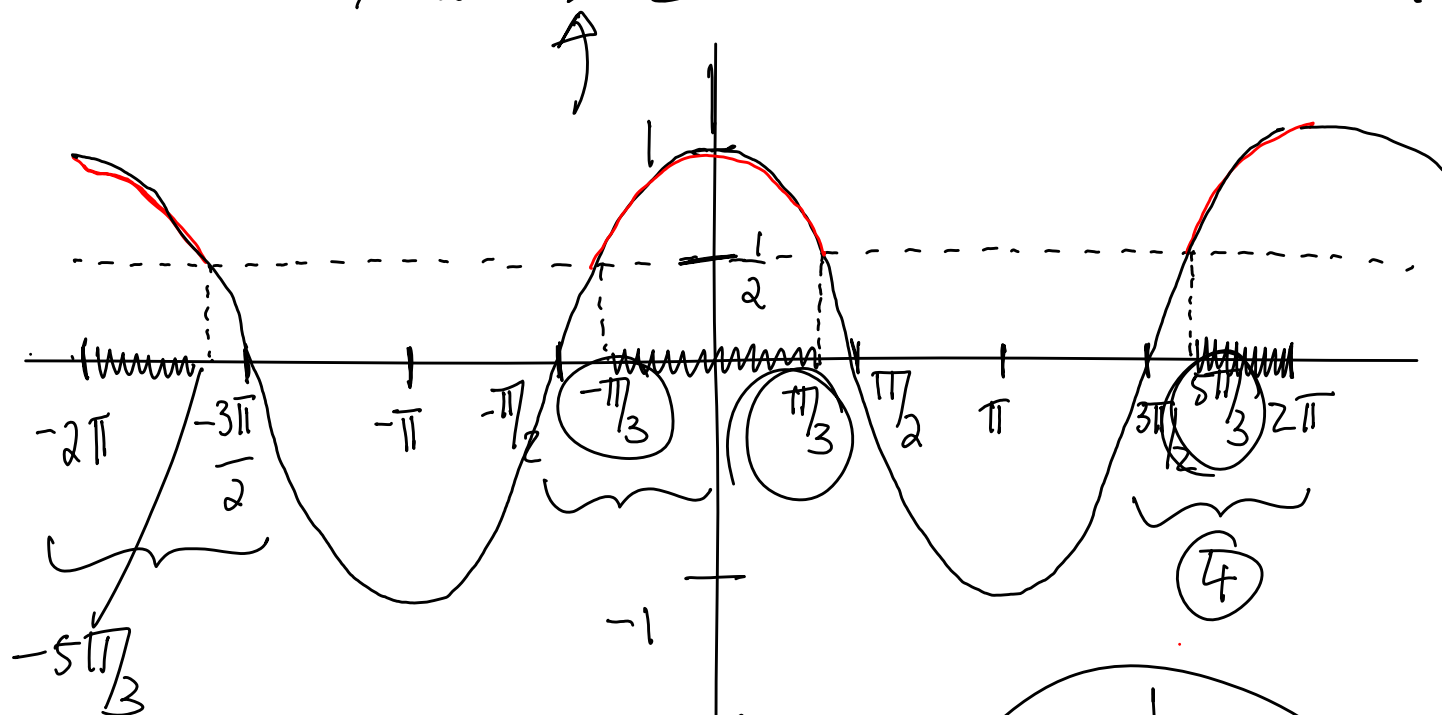
$$y = \cos x$$



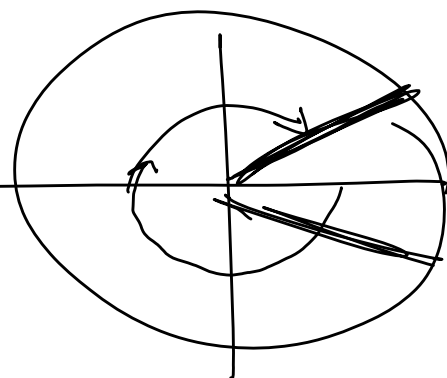


Solving trig inequalities

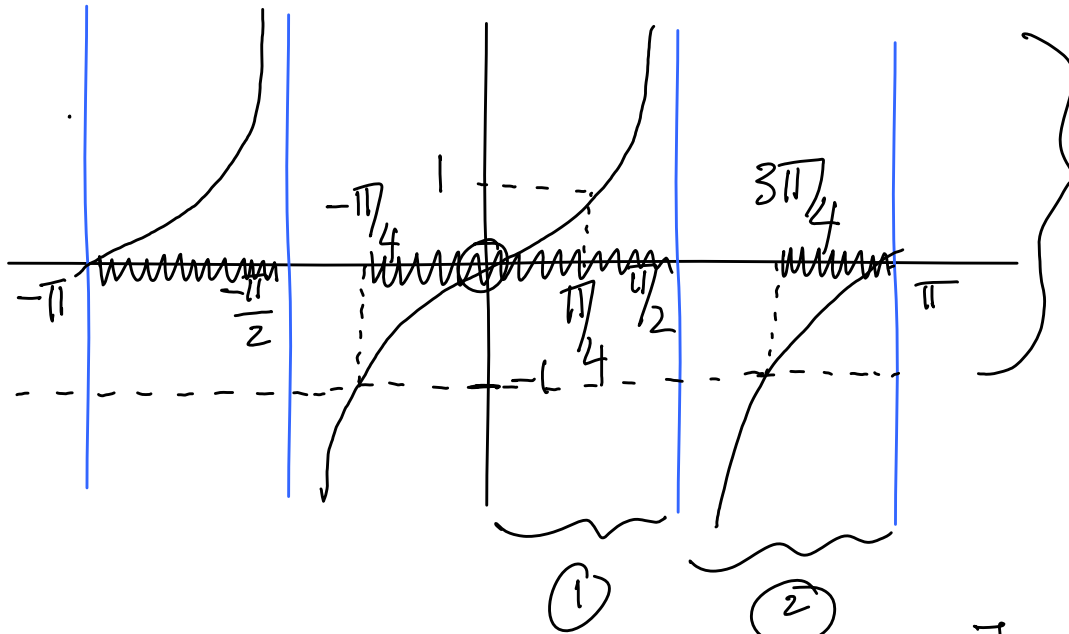
1. Solve $2 \cos x - 1 > 0$ for $x \in [-2\pi, 2\pi]$
 $\Rightarrow \cos x > \frac{1}{2}$



$$x \in [-2\pi, -\frac{5\pi}{3}) \cup (-\frac{\pi}{3}, \frac{\pi}{3}) \cup (\frac{5\pi}{3}, 2\pi]$$



2. Solve $\tan x > -1$, $x \in [-\pi, \pi]$



$$x \in [-\pi, -\pi/2) \cup (-\pi/4, \pi/2) \cup (3\pi/4, \pi].$$

